

Percent Applications: Tax, Tip, Discount, and Markup

Answer Key

Grades 6–7 Ratios and Proportional Relationships

Quick Answers

1. 2.4	4. 108	7. 57.24	10. 72
2. 5.4	5. 26.28	8. 26.97	
3. 36	6. 68.91	9. 40	

Common wrong answers

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5. If they got 1.78: found the tax but reported it as the total, forgetting to add the price back
6. If they got 10.51: computed the tip alone and called it the amount paid
7. If they got 58.32: combined the two percents into one 19 percent change instead of applying them one after the other to different bases
8. If they got 8.37: took 45 percent of the wholesale cost as the retail price instead of adding the markup to the cost
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- 1.** Jacket \$40, sales tax 6% — find the tax in dollars.

The base is the price, \$40, because the tax is charged *on the price*. “6% of” means multiply by $\frac{6}{100}$: $40 \cdot \frac{6}{100} = \frac{240}{100} = 2.4$. The tax is \$2.40. 2.40

Grader’s note: the question asks for the tax only. A student who wrote 42.40 answered the total instead — correct method, wrong quantity.

- 2.** Dinner bill \$36, tip 15% — find the tip in dollars.

The base is the bill before the tip, \$36. $36 \cdot \frac{15}{100} = \frac{540}{100} = 5.4$, so the tip is \$5.40. 5.40

Mental-math check: 10% of 36 is 3.60 and 5% is half of that, 1.80; $3.60 + 1.80 = 5.40$ ✓

- 3.** Backpack \$45, 20% off — ratio table and sale price.

Completed table: 100% of the price is \$45; the sale is $100 - 20 = 80\%$ of the price. Because percents are proportional to dollars, 1% is $\frac{45}{100} = 0.45$, so 80% is $80 \cdot 0.45 = 36$.

In one step, multiply by the *remaining* fraction: $45 \cdot \frac{100 - 20}{100} = 45 \cdot 0.8 = 36$. The sale price is \$36.00. 36.00

- 4.** Bike cost \$80, markup 35% — find the selling price in dollars.

Markup is added to the shop’s cost, so the base is \$80 and the multiplier is $1 + \frac{35}{100} = 1.35$.

$80 \cdot 1.35 = 108$, so the shop sells it for \$108.00. 108.00

Structure to point out: the markup alone is $80 \cdot 0.35 = 28$ dollars, and $80 + 28 = 108$ — the same answer in two steps instead of one.

5. Calculator \$24.50, sales tax 7.25% — find the total paid.

Tax is added to the price, so multiply by $1 + \frac{7.25}{100} = 1.0725$: $24.50 \cdot 1.0725 = 26.27625$, which rounds to \$26.28 at the nearest cent. 26.28

Grader's note: the tax by itself is $24.50 \cdot 0.0725 = 1.77625 \approx 1.78$ dollars. A student who wrote 1.78 found the tax and forgot to add the price back.

6. Restaurant bill \$58.40, tip 18% — find the total paid.

The tip is added to the bill, so the multiplier is $1 + \frac{18}{100} = 1.18$: $58.40 \cdot 1.18 = 68.912$, which rounds to \$68.91. 68.91

Grader's note: 10.51 is the tip alone ($58.40 \cdot 0.18 = 10.512$), not the amount paid.

7. Coat \$72, 25% off, then 6% sales tax on the sale price.

Step 1 — discount. Base is the original price: $72 \cdot \frac{100 - 25}{100} = 72 \cdot 0.75 = 54$ dollars.

Step 2 — tax. The base changes: tax is charged on the *sale* price, so $54 \cdot 1.06 = 57.24$ dollars. 57.24

Grader's note: 58.32 is what you get by folding the two percents into a single 19% change ($72 \cdot 0.81$). That is wrong because the two percents act on different bases — the discount on \$72, the tax on \$54.

8. Board game cost \$18.60, markup 45% — find the shelf price.

Markup is added to cost, so multiply by $1 + \frac{45}{100} = 1.45$: $18.60 \cdot 1.45 = 26.97$ dollars. 26.97

Grader's note: 8.37 is 45% of the cost ($18.60 \cdot 0.45$) — the markup amount, not the shelf price. Add it to the cost: $18.60 + 8.37 = 26.97$ ✓

9. Sweater sells for \$34 after 15% off — find the original price.

The discount is taken off the *original* price, so the original price is the base and \$34 is $100 - 15 = 85\%$ of it. With x the original price in dollars:

$$x \cdot \frac{100 - 15}{100} = 34 \quad \implies \quad 0.85x = 34 \quad \implies \quad x = \frac{34}{0.85}$$

So the original price was \$40.00. x = 40.00

Check: 15% of 40 is 6 dollars, and $40 - 6 = 34$ ✓ Taking 15% off the \$34 sale price instead is the classic error — it uses the wrong base and gives about 28.90.

10. Receipt total \$77.04 including 7% tax — find the price before tax.

Tax is charged on the pre-tax price, so that price is the base and the receipt total is 107% of it. With x the price before tax in dollars:

$$x \cdot \left(1 + \frac{7}{100}\right) = 77.04 \quad \implies \quad 1.07x = 77.04 \quad \implies \quad x = \frac{77.04}{1.07}$$

So the price before tax was \$72.00. x = 72.00

Check: $72 \cdot 0.07 = 5.04$ dollars of tax, and $72 + 5.04 = 77.04$ ✓ Subtracting 7% of the total ($77.04 \cdot 0.07 = 5.3928$) uses the wrong base and misses by about 0.35.